

Zhuoran Yu

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Education

Ph.D. in Computer Science

Sep 2021 – May 2026

University of Wisconsin–Madison, Madison, WI

Advisor: Yong Jae Lee

M.S. in Computer Science

Aug 2018 – Aug 2020

Georgia Institute of Technology, Atlanta, GA

GPA: 4.0/4.0

B.Math in Computer Science and Applied Mathematics

Jan 2014 – Jun 2018

University of Waterloo, Waterloo, Canada

Graduated with Distinction & Dean's Honor List

Arthur Beaumont Memorial Scholarship

Research and Professional Experience

Senior AI Research Scientist

Jun 2026 – Present

Autodesk, Inc., San Francisco, CA

- Conduct research and development at Autodesk Research AI Lab, focusing on foundation models for 3D content generation and multimodal design intelligence.
- Develop and evaluate generative AI methods for three-dimensional design and shape generation from multimodal inputs.

Research Scientist Intern

May 2025 – Aug 2025

IBM Watson AI Lab, Cambridge, MA

- Contributed to the development and evaluation of Granite vision-language models, focusing on document and multimodal understanding.

Research Scientist Intern

May 2024 – Jan 2025

Meta FAIR, Language and Communication Team, Seattle, WA / Remote

- Conducted research on data mixture strategies for early-fusion multimodal large language models.

Research Scientist Intern

May 2023 – Nov 2023

Meta, CORE AI Team, Menlo Park, CA / Remote

- Developed synthetic-data-based approaches for improving visual recognition models using pre-trained diffusion models.

Applied Scientist Intern

May 2022 – Aug 2022

AWS AI Rekognition, Seattle, WA

- Developed semi-supervised learning methods for 2D human pose estimation using denoised pseudo-heatmaps and uncertainty-guided pseudo-label selection.

Publications

* denotes equal contribution

DocHop: Benchmarking Out-of-domain Multi-hop Reasoning in Information-Dense Documents

Zhuoran Yu, Le Thien Phuc Nguyen, Jaden Park, Xinyi Gu, Zexue He, Soochahn Lee, Rogerio Feris,

Yong Jae Lee

International Conference on Machine Learning (ICML), Seoul, South Korea, 2026.

Large Language Model Teaches Visual Students: Cross-Modality Transfer of Fine-Grained Conceptual Knowledge

Thomas Shih-Chao Liang*, **Zhuoran Yu***, Yong Jae Lee

International Conference on Machine Learning (ICML), Seoul, South Korea, 2026.

Revisiting Active Speaker Detection: An In-the-Wild Benchmark for Generalization and Robustness

Le Thien Phuc Nguyen*, **Zhuoran Yu***, Khoa Quang Nhat Cao, Yuwei Guo, Tu Ho Manh Pham, Tuan Tai Nguyen, Toan Ngo Duc Vo, Lucas Poon, Tuan Khai Nguyen, Soochahn Lee, Yong Jae Lee
INTERSPEECH, Sydney, Australia, 2026.

See, Hear, and Understand: Benchmarking Audiovisual Human Speech Understanding in Multimodal Large Language Models

Le Thien Phuc Nguyen*, **Zhuoran Yu***, Samuel Low Yu Hang, Subin An, Jeongik Lee, Yohan Ban, SeungEun Chung, Thanh-Huy Nguyen, JuWan Maeng, Soochahn Lee, Yong Jae Lee

Findings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Findings), Denver, Colorado, 2026.

LASER: Lip Landmark Assisted Speaker Detection for Robustness

Le Thien Phuc Nguyen*, **Zhuoran Yu***, Yong Jae Lee

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV Oral), Tucson, Arizona, 2026. .

Composition-Grounded Instruction Synthesis for Visual Reasoning

Xinyi Gu, Jiayuan Mao, Zhang-Wei Hong, **Zhuoran Yu**, Pengyuan Li, Dhiraj Joshi, Rogerio Feris, Zexue He

International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, 2026.

DAVE: A VLM Vision Encoder for Document Understanding and Web Agents

Brandon Huang, Hang Hua, **Zhuoran Yu**, Trevor Darrell, Rogerio Feris, Roei Herzig

International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, 2026.

How Multimodal LLMs Solve Image Tasks: A Lens on Visual Grounding, Task Reasoning, and Answer Decoding

Zhuoran Yu, Yong Jae Lee

Conference on Language Modeling (COLM), Montreal, Canada, 2025.

CuRe: Cultural Gaps in the Long Tail of Text-to-Image Systems

Aniket Rege, Zinnia Nie, Mahesh Ramesh, Unmesh Raskar, **Zhuoran Yu**, Aditya Kusupati, Yong Jae Lee, Ramya Korlakai Vinayak

IEEE/CVF International Conference on Computer Vision (ICCV), Honolulu, Hawaii, 2025.

Diversify, Don't Fine-Tune: Scaling Up Visual Recognition Training with Synthetic Images

Zhuoran Yu, Chenchen Zhu, Sean Culatana, Raghuraman Krishnamoorthi, Fanyi Xiao, Yong Jae Lee

Transactions on Machine Learning Research (TMLR Featured Certificate), 2025.

Denoising and Selecting Pseudo-Heatmaps for Semi-Supervised Human Pose Estimation

Zhuoran Yu*, Manchen Wang*, Yanbei Chen, Paolo Favaro, Davide Modolo

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Waikoloa, Hawaii, 2024.

InPL: Pseudo-labeling the Inliers First for Imbalanced Semi-Supervised Learning

Zhuoran Yu, Yin Li, Yong Jae Lee

International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023.

Group R-CNN for Weakly Semi-supervised Object Detection with Points

Shilong Zhang*, **Zhuoran Yu***, Liyang Liu*, Xinjiang Wang, Aojun Zhou, Kai Chen

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, 2022.

Scale-Equalizing Pyramid Convolution for Object Detection

Xinjiang Wang*, Shilong Zhang*, **Zhuoran Yu**, Litong Feng, Wei Zhang

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Virtual, 2020.